Chapter 5
Service Management Overview

Introduction

Service Management is designed to operate independently or as an integrated part of other modules.

The package will support multi-branch operations within each of a number of companies.

Contract maintenance is supported by quotations, pricing and renewal to the production of maintenance invoices, including in advance, arrears or by instalment.

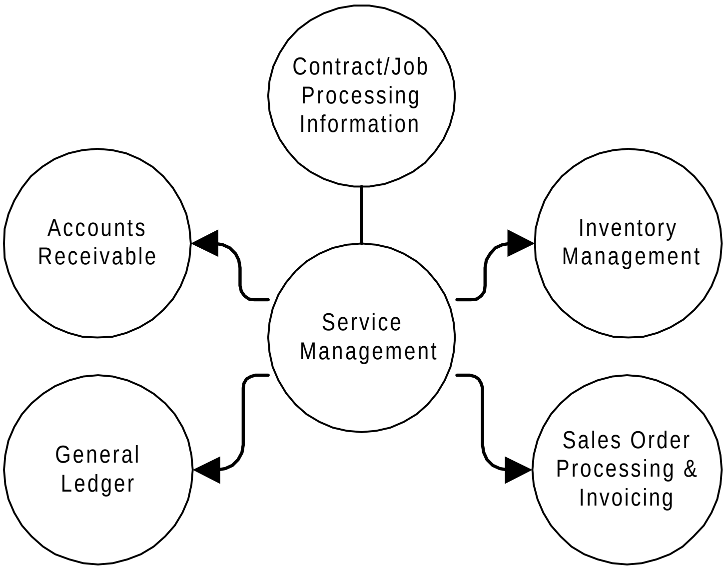
Job processing is controlled by service call logging, engineer allocation and planned maintenance. Job escalation facilities are also provided. Full enquiry features enable access to master files, transaction history, job status, and financial data.

Where System Manager is installed, security is provided by restricting access to certain functions by operator ID, or by creating tailor-made menus for each operation. This means that operators can be prevented from accessing specific functions, or information belonging to certain companies.

Relationship to Other Modules

Service Management operates under the control of System Manager. The full benefits of the system are gained by integrating it with the following prerequisite modules:

- Accounts Receivable
- General Ledger
- Inventory Management
- Sales Order Processing



You can also optionally integrate it with the following:

- Advanced Financial Integrator (AFI)
- Distribution Requirements Planning
- Customer Returns
- World Trade
- Forecasting

Module Configuration

As with all modules, Service Management can be operated for a number of companies, the characteristics of each being maintained on a control file. The module is controlled by setting up data for the company and then for each branch within the company. This would normally be done after the Accounts Receivable, Inventory Management and Sales Order Processing companies have been created.

Accounts Receivable, Inventory Management, Sales Order Processing and Service Management modules all use the same company codes. You can add branches after starting the module.

Certain data and policies are defined at the company level. These include:

- Document reference numbers: invoice, credit note, contract, job
- General Ledger accounts
- Default labour price list
- Default scheduled visit profile
- Working day start / finish times
- Hours entry only, for call reporting.

Additionally, the details of service and accounting periods are defined at company level. Up to 99 service periods per year are allowed, and are used mainly for the scheduling of planned maintenance visits. Each day of the year may be defined as a working or non-working day, to enable calculation of target response date / time for service jobs.

Up to 99 accounting periods per year are allowed and these should be consistent with those specified in the ledgers. The number of days in the accounting periods is used to enable the apportionment of deferred revenue from service contracts invoiced in advance.

Once the control parameters for a company have been defined then at least one branch will be defined. A branch is a physical or logical location that has responsibility for servicing specific customer sites.

Certain data and policies are defined at the branch level. These include:

- Branch name and address
- Printer output queue for job audit log
- Job escalation parameters: escalation time fence
- Escalation reporting interval
- Users to be alerted at each escalation level
- Default response hours.

Each user must be authorised to access at least one branch within a company, but may be authorised to several or all branches if required. This authorisation by branch will restrict the data to which a user has access.

Module Reference Data

Service Management uses a number of reference files that hold data which rarely changes, as follows:

- Standard codes and descriptions
- System parameters
- Customers
- Divisions, model groups, model sub-groups and models
- Installations
- Contract types
- Job categories
- Labour rate price lists
- Service territories, teams and field service groups
- Engineers
- Contract rates
- Scheduled visit profiles

Standard Codes and Descriptions

A descriptions, or codes / parameters file enables codes to be set up with standard descriptions. This provides for validation at the time of data entry and also the display of descriptions on both screens and reports. In some specific instances a parameter or number is also stored, as for example in the case of value added tax percentages.

Certain standard codes and parameters must be present for the successful operation of Service Management, and these are supplied with the system. Examples of code types and values that might be set up are as follows:

Code Type	**Type Description**	**Code Value**	**Code Description**
FLTC	Fault Code	OVH ELE	Overheating Electrical Fault
EGRD	Engineer Grade	GR1 GR2	Grade 1 Grade 2
LHTY	Labour Hours Types	BASE DBLE	Basic Double Time

Setting Up Customers and Equipment

Before contract information can be recorded, or service operations entered, customer details, the location of customer sites and the equipment installed at these sites must be entered.

The set of procedures described in this section enables you to set up and maintain:

- Customer accounts, maintained through Accounts Receivable.
- Installation addresses, maintained through Accounts Receivable.
- Installed equipment details.
- Equipment models.
- Model group and sub-group maintenance.

Note that some of this information is also used by Accounts Receivable and Sales Order Processing. Care must be exercised in its maintenance to make sure that data is correctly held for all purposes.

Each set of information can be printed or used for enquiry purposes.

Before starting to set up customer and installation details in Service Management you have to use the Names And Addresses option in Accounts Receivable and the Installation Address Maintenance, in Accounts Receivable and Sales Order Processing.

Names and Addresses Maintenance

A customer account holds the customer's name and address, plus accounts-related data, such as payment terms and credit limit.

Once an account is defined, the installation site addresses for that customer can be maintained. The account address can itself be an installation site.

Additional service details are linked to account and installation addresses. Three maintenance screens are involved. The routine for maintaining these records is described in the Additional Service Details Maintenance section of Service Management.

No installation records or contracts can be entered, and no transaction processing can be performed, without customer accounts. Although some of the data held on the customer account is not used by Service Management, the following parameters are significant:

- Name and address - Displayed on many screens to allow verification of correct customer account. Printed on some reports, including invoices / credit notes.
- Geocode or postcode - Used in automatically assigning teams of engineers to jobs at the customer site, by means of the geocode / territory definitions and engineer team / field service group / territory assignment rules.

- VAT code & indicator - Used within invoice / credit note generation to establish which VAT rate to charge.
- Credit stop indicator - Used within job maintenance to provide warning that the customer is on credit stop.

Installation Address Maintenance

This option is used to maintain customer site addresses where equipment is installed.

An installation address is a customer address at which equipment is installed, which may require service. A customer account address may itself be an installation address, and there may be several additional installation addresses for the customer account.

When you set up a customer account, a three digit suffix (address code) of 000 is appended. Further installation addresses are then created through this option by assigning address codes of 001, 002, etc. to each address record for the account. The codes can be numerics or characters, as preferred.

Each installation address is identified by a unique combination of account and address code.

Additional service details are linked to account and installation addresses. Three maintenance screens are involved. The routine for maintaining these records is described in the Additional Service Details Maintenance section.

The site address must be created before the details of equipment installed at the address can be entered. No contract or transaction processing can be performed until this equipment database is present. The following data for an installation address is significant:

- Name and address - Displayed on many screens to allow verification of correct customer address. Printed on some reports, including invoices / credit notes.
- Postcode - Used in automatically assigning engineers to jobs at the customer site, by means of the postcode, or geocode / territory definitions and engineer / territory assignment rules.

Price List Options

The two price list options are both Sales Order Processing options, so turn to SOP product information for more details regarding these.

Customers

Service Management shares a common customer database with Accounts Receivable and Sales Order Processing, with maintenance routines in each module reflecting its specific requirements. For example, Accounts Receivable controls credit limits and payment terms, Sales Order Processing controls stock allocation, pricing of sold items and Sales Analysis coding, and Service Management controls pricing of service items and invoice consolidation levels.

The structure of the customer information can be manipulated to match the structure of the customer's organisation. On the one hand there may be a simple single account, on the other a statement account with many 'invoice to accounts' each with many different delivery or installation accounts.

Statement Account

This is shared with Accounts Receivable and is normally created and maintained by it. The data includes credit details, Accounts Receivable parameters and statistical codes.

Invoice Account

Invoices may be sent to an invoice only address.

Delivery or Installation Account

Each account may have many delivery or installation addresses. These are defined by Sales Order Processing and / or Service Management, by adding a three-character address code suffix to the account code.

Additional Information

Each account address, at which equipment is to be serviced, must have additional information set up, which is used solely by Service Management. The information includes the invoice destination address, invoice consolidation level (account, address code, or contract / job), service region, and price list for service parts.

Identification of a customer can be made either by entering the account and address code, or by using the powerful name and address search routines that are available in the system.

Models

Each piece of equipment defined to Service Management is identified by its model code and, normally, its serial number. Standard model codes / numbers are used to describe the different types / configurations of equipment that can be installed. Each model code is linked firstly to a model sub-group which in turn is linked to a model group.

Models may be specified as individual machines, or as peripherals to a machine.

Contract rates for different types of service contract can be held at model group, model sub-group or model code level; engineers can be assigned to perform work only on specific model groups, model sub-groups or models within a service territory.

The labour rates to be charged for service work are determined by the model on which the work is performed, since each model code holds a pointer to its appropriate labour price list.

Installations

An installation within Service Management can be defined as a customer address at which equipment or systems are installed.

The customer address is identified by an account code and address code and can have many items of equipment or systems installed. This configuration of equipment or systems at a customer location constitutes an installation record. Records should be set up for all installations irrespective of whether they are covered by service contracts.

Each piece of equipment at an installation is identified by the unique combination of its model number and serial number. If serial numbers are not known, then the model number and model quantity can be used.

The data held for each piece of equipment at an installation includes details of age, dates of delivery and installation, warranty period, location / department, complexity points and product division.

Engineers

Each engineer who will carry out service / repair work is defined on the engineer master file. Details include his name and address, radio page number, team code, grade, hourly cost rate, service region and stockroom code, together with his skills matrix.

The stockroom code provides the link to Inventory Management and must be defined therein. It represents a discrete area within a company where stock is recorded and controlled separately from other company stocks. In the case of service engineers, the stockroom could be the specific van or car of the engineer himself, or a common store or depot from which spare parts are drawn.

Service Territories And Engineer Assignment

Service Management facilitates the automatic assignment of engineer teams to service jobs, provided that certain details have been defined:

- The customer records must hold the geographic or postcode of the customer address.
- The geographic area covered by the company must be divided into service territories; each territory will be defined by a list of full or partial geocodes / postcodes.
- The engineers' teams who cover each territory must be specified.
- The field service group for the team must be defined, to show which product divisions, model groups, model sub groups and models the team will service. The field service group record contains a skills matrix to define these limits.

The system can map the customer's address to a service territory and allocate work to the team of engineers designated to operate in that territory.

The allocation of an individual engineer within the team, to a specific job, can be handled automatically or manually. Further specialisation is used in job selection according to the engineer's skill level, i.e. an engineer can be assigned to service specific model groups (and optionally sub-groups) and / or to perform work of a specific type (i.e. job category), and / or to perform work on equipment belonging to a specific division.

Contract Types

Contract types are used to define the different types of service contract offered by the company to its customers. Along with its description, each contract type record holds the normal duration of the contract, guaranteed response hours, invoice schedule (in advance, in arrears, or on completion of each scheduled maintenance visit), and whether a quotation or contract is generated at renewal time.

Billing parameters, which may be applied to the contract type, are also selected (i.e. to indicate if service charges, rental charges, or various meterage conditions will apply).

All contracts created in the system will have a user-defined contract type code. The system will default a contract type code of *NO for work on equipment not covered by a service contract.

The parameters set up for the contract type will act as defaults for the contract, but can be changed if required.

Job Categories

Job category codes are used to describe the different types of service work to be performed. A number of categories can be defined as system defaults for breakdowns, warranty and repeat calls etc, to save time at call logging.

Additionally, a table of valid job category codes for each contract type must be defined. This table is used to specify whether the customer should be charged for travel and labour hours (both fixed and hourly charges), for up to four parts groups and for miscellaneous charges, when work of a certain job category is carried out on equipment covered by the specific contract type.

Price Lists

Two types of price list are used by Service Management, for labour rates and for parts.

Labour rate price lists define the hourly labour rates to be charged to customers, where appropriate, for contract, non-contract and workshop work. The price lists include an hourly travel rate; they carry effectivity dates, to control the module of new prices, and are linked to models: the model code of the equipment being serviced / repaired determines the labour rates to be charged.

Parts price lists and discount lists define the prices and discounts to be applied when charging for spare parts fitted or used by an engineer. These lists, which also carry effectivity dates, are maintained in Sales Order Processing, but are linked to the customer additional details records in Service Management: the prices charged for service parts are determined by the customer site at which the work is done.

Contract Rates

Contract rates are defined for each contract type at model group, model sub-group, or individual model code level.

Contract rates specify the charge per scheduled maintenance visit and / or a fixed charge for each piece of equipment covered by the contract type. Contract rates carry effectivity dates to control the module of new rates, and are used by the contract pricing routines in Service Management.

Contract rates can only be used when the system has been defined for term processing; they cannot be used for monthly processing.

Scheduled Visit Profiles

Scheduled visit profiles are used to define standard schedules of planned maintenance / service visits to equipment under contract. The schedules are defined in terms of the number, type and frequency of visits.

A visit profile can be associated with a contract, model, or model group, and enables the automatic generation of the service visit schedule for any piece of equipment added to a service contract.

Contract Processing

Service Management provides flexible and comprehensive facilities for contract management, as follows:

- Maintenance
- Pricing and invoicing
- Renewal
- Quotations
- Credits

Contract Maintenance

Service contracts are identified by a combination of contract number, type and start date. Each contract relates to one or more locations of a single customer account, and may cover one or more pieces of equipment installed at these locations.

When a new contract is entered on to the system, the contract number, type, start date, customer account and address code are specified. Contract header and equipment records are created.

The contract header record includes the contract duration and end date, order reference and date, and invoice destination. The service parameters include response hours, scheduled visit profile, allow service cover while the contract is pending, and the rates to be applied. The billing parameters set the invoice term, invoicing frequency and whether charges are in advance, in arrears or per visit. The service conditions provide for setting up service, visit and rental charges at contract header level: the service and visit charges override the contract rates set by model group, model sub-group or model; the service conditions can be overridden at equipment and peripheral levels.

Once you have set up the header you must define the contract equipment. The system lists all the equipment defined for locations of that account, and you can select those items to be placed on the new contract. Equipment placed on a contract must have been previously specified as resident at a location of the account to which the contract applies.

Service conditions at equipment and peripheral levels can be added to the contract. You can create a schedule of planned service visits, by service period for the duration of the contract, for each item on a contract. This can be done automatically by the system using the service visit profile associated with the contract header, model, or model group, but you can amend the schedule if required.

Contract Pricing And Invoicing

Contracts which are invoiced in advance or in arrears are priced and invoiced as follows:

Service Pricing

- A contract is priced for a defined period of time (invoice term) which may be the same as the contract duration itself, or may be less. The charge for an invoice term can in turn be split into instalments for invoicing purposes, for example a one year term can be invoiced in quarterly instalments. Additionally, the system will pro-rata the charge for equipment added to a contract part-way through its invoice term.
- To price a contract, each piece of equipment covered is priced individually and accumulated to give a total contract price. If the piece of equipment and its peripherals have special prices held, then these are the prices to charge.
- If no special price exists, then the contract rates file is accessed (term processing only), to obtain the price per maintenance visit and any supplement fee. The total number of maintenance visits scheduled for the piece of equipment in the invoice term is then multiplied by the price per visit and the supplement fee added. The result is the gross charge for the piece of equipment.

Note: Service contracts may include fixed visit charges, to be invoiced after each visit: the contract's billing parameters may be set accordingly. Such visits are invoiced through job invoicing at call completion, not through contract invoicing.

Rental Pricing

- A rental charge may be made for equipment on contract (if rental is defined on the contract type). The charge may be applied on the contract header, in which case it will be used as the default price for each item added to the contract, or may be maintained on the contract equipment line. The charge entered will be per month if monthly processing is selected on the system parameter file, or per term if term processing has been selected.

Meterage Pricing

- The following meterage charges may be applied (if selected on the contract type). Meterage charges can only be used with 'monthly processing'.
- A minimum volume can be defined on the contract header or equipment line. The minimum volume will be multiplied by the price per copy, to reach a minimum monthly charge, which may be invoiced in single or multiple months.
- A minimum charge can be defined on the contract header or equipment line, which will be the minimum monthly charge; this may be invoiced in single or multiple months.

- Up to five copy charge bands, for each of two meters, can be defined for each contract equipment line.
- Interim pricing allows for charges to be made according to a meter estimate, or actual reading returned by the user, based on the minimum units already charged for, (or actual units used if no minimum has been entered). The frequency of interim pricing is defined on the contract header.
- Reconciliation pricing allows for charges to be made according to an actual meter reading and allows for additional meterage charges to be applied, or credits given for overcharges. The reconciliation term must be a multiple of the interim period.
- Meterage pricing can be 'pooled', i.e. the minimum volumes / values can be spread across all of the equipment on the contract, allowing the excess units on one machine to be offset against the under-usage on another machine.

Invoicing

- Contract pricing creates records on an invoices pending file. A second function, generate and print invoices, generates invoice header and line records. At this stage, invoice consolidation, the grouping of invoice pending lines into invoices, takes place as specified on the customer additional details records, i.e. by account, account / location or contract. VAT is calculated according to the parameters set on the customer account record. The invoice documents are then printed.
- Invoice generation posts the relevant transactions to the sales and general ledgers. Where advance invoicing is performed, the revenue (i.e. income) value of the invoice is deferred in the General Ledger, by generating additional postings which 'transfer' period revenue values from a deferred income account into a sales account each period.

Contract Invoice Term

When pricing a contract, the price is calculated to cover a fixed period of time known as the contract invoice term. There can be one or many invoice terms within the life (duration) of a contract.

The contract invoice term is defined as the period of time from the next term starts on date to the next term ends on date, and defines the period of price protection for the contract.

Next Term Starts On Date

The date on which the next contract invoice term to be processed will start. Contracts are selected for pricing if the next term starts on is less than or equal to the date entered on the price contracts prompt screen. This field is held and maintained on the billing parameters window.

Next Invoice Term Ends On Date

The date on which the next contract invoice term to be processed will end. This field is held and maintained on the billing parameters window.

Normal Invoice Term

The length of time in months to be added to the next term starts on date, to calculate the subsequent next term ends on date and hence define the normal invoice term for the contract. This field is held and maintained on the billing parameters window.

Invoice Frequency

A sub-division of the normal invoice term. This designates the period of time, in months, between invoices, if the invoice term of the contract is to be invoiced in instalments. This field is held and maintained on the billing parameters window.

Contract Payable in Advance

A contract for which contract charges for any contract invoice term, or instalment of a term, are invoiced before the term or instalment.

Contract Payable in Arrears

A contract for which contract charges for any contract invoice term, or instalment of a term, are invoiced after the term or instalment.

Contract Payable Per Visit

A contract for which contract charges are invoiced as a result of a scheduled service visit being made and technically reported. The pricing of these visits is actually carried out by the Price Jobs option.

Pending Invoice Line

An invoice line, not yet assigned an invoice number or posted to the Ledgers, which is waiting to be grouped with other invoice lines to form an invoice. This grouping is dependent on the invoice consolidation level, which is set up for the customer account (i.e. the 000 record) by the Additional Service Details file maintenance option.

Note: The account code (000) sets the invoice consolidation level for all sites.

Pending Invoice Line Availability Date

The date on, or after which, the pending invoice line may be included in an invoice. For advance contracts, this will be the start date of the contract invoice term or instalment period; for arrears contracts, this will be the end date of the contract invoice term or instalment period.

Description

You can select all branches or a single branch, specify the type of billing, and enter a cut-off date relating to the contract invoice term(s). Select **F8=Submit** to submit a batch job.

Only contracts payable in advance or arrears are processed by this function; visit only contracts are excluded.

Pending invoice lines are generated for:

- All non-invoiced pieces of equipment, added to a contract during an invoice term, which have been priced previously.
- All equipment on contracts where the entered cut-off date is later than, or equal to, the next term starts on date of the contract.

Pricing Routine

Contract pricing is governed by the conditions set up in contract maintenance, at header or equipment levels, or at both.

Default Prices from the Contract Rates File

Note: Default pricing can only be used for fixed service, rental and visit charges. It will not apply to copy / vend contracts, where special contract conditions must be set up.

If the service conditions, at contract header or contract equipment levels for the fixed service, rental or visit charges, are not overwritten, their values default from the contract rates file. Contract pricing picks up the fixed service, rental and visit charges from the contract rates file, for the appropriate effective date, equipment group, contract type, term and currency.

For a particular currency, the contract rates record used will be that for the contract type and term, effective on the 'pricing date'. The 'pricing date' is derived as follows: if the 'use start date rates' field on the contract header is set to '1=Yes', or if there is a 'price held until date' which is later than the 'next term starts on date', the pricing date is the contract start date; otherwise, the pricing date is the 'next term starts on date'.

The equipment model group, model sub-group and model are compared to the same fields on the contract rates record; the record with the best match will be selected. If no record is found, the contract could be left with no charge: special service conditions could be set up by the user for the contract. The recommendation is that records should be created on the contract rates file for as many combinations of equipment and contract types as are necessary.

The price is calculated, by multiplying the visit charge by the number of scheduled maintenance visits for this piece of equipment which occur in the 'contract invoice term': between the 'next term starts on' and the 'next term ends on' dates.

The fixed service charge, from the contract rates record, is added; the charge is deemed to apply to the 'normal invoice term', and is adjusted to allow for the 'contract invoice term': the number of days in the 'contract invoice term' is divided by the number of days in the 'normal invoice term', and this factor is applied to the fixed service charge.

The rental for the contract term is included, but separately from the service and visit charges.

Contract Conditions Prices at Header / Equipment Levels

If a price is keyed into the Contract Conditions window at header or equipment levels, this price is deemed to apply to the 'normal invoice term' specified on the contract. The 'normal invoice term' is converted to days, by adding the 'normal invoice term' in months to the 'next term starts on' date, to get the 'normal invoice term ends on date', and to work out the number of days included between these two dates. The number of days between the 'next term starts on' date and the 'next term ends on' date (from the contract header) is also calculated. The ratio, of the latter number of days to the former, is calculated and the same ratio is applied to the contract conditions price, to get the actual price for the contract, or for the specific piece of equipment, for the contract invoice term.

Note that for complete instalment periods, the invoices are calculated to be of equal value.

Pro-Rata Pricing

The price for equipment added late or removed early from a contract will be calculated on a pro-rata basis, as follows:

The number of days in the 'normal invoice term' of the contract is calculated, as described above.

For equipment added late, the contract invoice term is the number of days between the date of addition of the piece of equipment to the contract, and the 'next term ends on' date, or the equipment removal date, whichever comes first.

For equipment removed early, the contract invoice term is the number of days between the date of addition of the piece of equipment, or the 'next term starts on' date, whichever comes last, and the 'next term ends on' date, or equipment removal date, whichever comes first.

The ratio of the two sets of days (i.e. contract invoice term and normal invoice term) to each other is calculated. This ratio is applied to the calculated price for the normal invoice term, to get the actual price for the contract, or for the specific piece of equipment, for the number of days that it has been on the contract.

Note: Equipment actually removed, or scheduled for removal, within an invoice term already processed, will not have an automatic credit generated; for this, use the Contract Credit option.

Contract Renewal

This function can be run at any time and prompts the user for a date representing the expiry date limit of contracts for renewal. The system reads through all active contracts and determines which ones to renew, by comparing the contract end date with the entered date. If the contract end date is less than or equal to the entered date, renewal is performed.

In order to renew a contract, the equipment on the contract must still be eligible for the contract type, i.e. the contract charge rates must be in effect.

A new set of contract records will be generated with either the same contract number or a new, system-assigned contract number depending on an indicator on the company profile. The contract start date for the new contract will be the day after the existing contract end date. Other contract details are copied to the new contract.

The status of new contracts will be set as either 'pending start date' or 'quotation', depending on the 'quotation required at contract renewal' flag on the contract type record. The user can, if required, amend the contract details generated (through contract maintenance procedures) before quotations or invoices are generated.

Contract Quotations

Quotations are created in the same way as contracts, but have a special status of 'quotation'. They may only be created for contracts with service and visit charges; rental and meterage contracts cannot have system-generated quotations.

Contract quotations can be produced on pre-printed stationery, to show account / site details, contract number, type, start date, duration, customer order number, equipment covered, number of scheduled maintenance visits and price. The quotation price is calculated in the same way as the contract invoice price.

Quotations can be converted to contracts, by using the contract maintenance facility to change the status from 'quotation' to 'pending contract'. A facility is also provided for generating quotations from existing contracts. This can be used, for example, if a customer requires a quotation for a different type of contract, covering the same equipment as his existing contract.

Contract Credits

Credits can be processed at any time, to refund part, or all, of a contract invoice, and to cancel any pending invoice records.

The contract, or the invoice, or the pending invoice period from / to is selected. The selected record(s) may be amended to affect the operative date and value(s) involved. The selected record may be discarded, a full or partial credit may be given, or the original charge may be credited and a different amount re-invoiced.

The resultant invoices and credit notes are produced when the option to Generate And Print (Contract) Invoices is next taken. Any cancelled pending invoice records are removed from the work file at the same time.

Job Processing

An important function of any service organisation is to complete, in an efficient and timely manner, both ad hoc customer service requests and planned maintenance work according to the contractual terms agreed.

On completion of service jobs, any chargeable work should be invoiced promptly in order to optimise cash flow.

Service Management provides the following facilities to enable these objectives to be met:

- Planned maintenance job generation
- Call logging
- Engineer work assignment and diary
- Engineer technical reporting
- Job pricing & invoicing
- Job escalation
- Remote communications links to and from engineers

Planned Maintenance Job Generation

This is a process that is run as required. Load PM Jobs creates a service job for each installation that has a planned maintenance visit scheduled for periods falling within a user-specified range. The function allows the schedule to be created for up to six periods in advance of the current period. The service schedule details, held for equipment covered by contract, provide the base data for processing.

The process will attempt to assign an engineer team to each service job, by comparing the customer geographic code with the territory definitions, and the model type requiring service with the field service group that covers the division, model group, model sub-group and model. If a specific engineer was entered against the equipment on the installation record, this engineer will always be assigned (if this method of selection is defined on the system parameter file, on screen 5).

If there is more than one piece of equipment at an installation address, on the same contract, requiring a service visit in the same period, then this procedure will normally generate only one job covering all the equipment. However, an option can be set on the company profile to generate an individual job for each piece of equipment.

Call Logging

This facility enables unscheduled / breakdown calls from customers to be recorded, and corresponding service jobs generated, for subsequent allocation to engineers. It also enables the details of existing jobs outstanding to be amended.

Addition of New Job

The logging of a new job requires entry of either the customer account code (or the account can be found using alpha search data), the service contract number, the serial number, or the model number / serial number. Any of these enables access to the customer installation record which holds the details of the equipment at the customer site. A new job can only be logged if an installation record exists for the customer address requesting service. A facility to create installation details for non-account customers is provided at this point.

If the customer is on credit hold / stop, a warning message is displayed and the call status is set to 'credit hold', but the job may still be entered.

If a contract number has been entered, all the equipment on the contract is displayed. If an account code and address code have been entered, all the equipment at the installation is displayed. Each item to be included on the new job can then be selected from the displayed list. If a specific model and serial number have been entered, the job details for the item are prompted for immediately.

If the customer has a number of items of equipment on site and cannot identify which of them requires the service, the job can be logged against any available machine line. When the engineer completes the job and reports on it, the true model / serial number must then be identified to the system.

The system will check for previous calls (for qualifying job categories), to determine if this is considered to be a repeat call: this compares the number of days and / or meter units which have elapsed since the previous call, with the values defined on the volume segment file for the model. The system allocates a job number; the account name and address is displayed and may be amended just for this job. Job line details are displayed, showing the contract / warranty cover; response hours are calculated by reference to the special serial numbers file, the contract line, the contract header, or the contract type (possibly reduced by the 3-D Matrix percentages). A target start date and time is set by adding the retrieved hours to the date and time the call is logged. The date and time the call is logged may be altered while the call is in 'Addition' mode, to allow for calls taken out of hours (e.g. by overnight answering service).

A history enquiry, showing details of previous jobs on the piece of equipment, is displayed in a window if the call is a repeat, or can be accessed by a function key. A customer order number may be entered; it is essential if requested on the customer additional details record and any elements of the call are chargeable.

The job category displayed for the call defaults to the value determined by the user in job category maintenance and may be amended. Customer contact name and fault code / description, for the item of equipment to be serviced / repaired under the new job number, are entered. The response hours can be altered if required, which will re-calculate the target response date / time for the job, which can then be monitored by the escalation procedure. Alternatively, if the customer requires an engineer to visit at a specific date / time, an appointment date and time can be entered. At this stage, an engineer can be assigned to the job, by displaying a selection window of the engineers in the team and their skills.

Amendment to an Existing Job

Access to existing service job records can be achieved through various routes – by entry of customer account code / address code, or serial number, or model / serial number, or contract number / type (if the customer has a service contract), or customer order number, or job number. Alternatively, the customer account can be accessed by specifying up to three search parameters (relating to pre-defined alpha search data fields from the account / address record).

A list of all current jobs on the system for the installation is displayed; the list includes both scheduled service and unscheduled / breakdown jobs. On selection of a job number, a list of all the equipment included on the job is displayed (a job may cover one, or more than one piece of equipment at an installation depending on the setting on the parameter file), along with job 'header' information, i.e. the job number, creation date and time, customer order number, customer contact, service contract number and type. On selection of an individual piece of equipment the details of job category, engineer assigned, customer contact, fault code, are displayed and can be amended as required.

Engineer Work Allocation

Engineer work allocation may be carried out by running the automatic call allocation program, or manually.

If automatic call allocation is required, the program is initiated by taking an option on the service operations menu. In addition, the Service Management sub-system must be running.

Note: The sub-system library list must be tailored to take account of:

- the use of consolidated files libraries in place of IN, OE, SL and G files libraries.

- the version of Business Manager in use, for adjusting the four IPG libraries.

If you are running System 21 Version 3 on an AS/400, then use the CHGJOBD command and **F4**, enter OSLSSP/OSLS2P3, then select **F10** and page down to the library list.

Automatic call allocation is a sleeper job, that will automatically schedule engineers within a team, to the jobs which have been assigned to the team in job logging. Assignment takes place on the basis of finding an available engineer, based on his current workload and skill profile, and who can achieve an estimated time of arrival that is equal to or less than the target date and time for the call. If the selection criteria cannot be met, an error message is displayed for the call, and a non-assignment reason code is entered on the call detail screen.

Calls can be scheduled manually, in which case the same criteria and checks for engineer selection are applied.

The function allows outstanding jobs to be despatched to service engineers, who phone or call in, to request work. It also enables entry of status updates from engineers on their current jobs, to indicate for example, that parts are awaited or the work is complete.

A list of jobs assigned to a specific engineer or team of engineers, or a list of all jobs for the branch can be displayed. Further selection parameters enable either planned jobs or unscheduled (call out) jobs, and jobs of certain status, to be extracted. The list of jobs displayed will be sequenced in order of target date / time, the most urgent jobs first.

Jobs for a team can be displayed graphically: colour blocks represent each job and the colour denotes the urgency of the job. Unscheduled jobs for the team are displayed in the lower part of the screen; scheduled jobs are shown under each engineer in the upper part of the screen. Jobs can be moved with the cursor and a function key. Job line details can be displayed, for team and engineer work queues, by positioning the cursor on any job and pressing 'Enter'.

The valid status codes for a piece of equipment on a job are:

Open;

Assigned;

Scheduled;

Despatched;

Work in Progress;

Requires Telephone Call;

Awaiting Parts;

Complete: Not Documented.

Status codes can be amended as required.

Engineer Technical Reporting

This is the facility used to enter the details of completed work from the engineer's reports, and is accessed by selecting the completed job line with 99.

A technical report is made up of four sections:

Header Details

Details of the engineer number, job number, visit date, report reference, time on site, customer and engineer travel time, are entered for each new technical report. Distance travelled can also be recorded and charged

Labour Hours

Labour hours spent on each piece of equipment are entered, qualified by fault code, job category, machine section and sub-section repaired, and corrective action. Hours can be booked at up to four different labour rates, i.e. basic plus three overtime rates. The hours are classified as non-chargeable (0%), or chargeable (100%), according to the charge matrix for the contract type / job category combination defined; the charge percentage can be changed to any intermediate value between 0% and 100%.

Parts Used

Spares parts used in carrying out the service or repair of each piece of equipment are entered, analysed by job category. For each part number, the quantity used is entered. The parts are classified as non-chargeable (0%), or chargeable (100%), according to the charge matrix for the contract type / job category combination defined. The charge percentage can be changed to any intermediate value between 0% and 100%.

Parts Ordering

The engineer uses this function to order spare parts to complete an existing job or a return job.

This reminds the engineer that parts have been shipped for the job when in the Call Reporting Parts Used Screen.

The engineer can order and use items that are not defined as being stocked in the relevant stockroom. The system creates the stockroom/item link automatically.

Service Management keeps track of the order for the job number. When a technical report is next entered, the parts are automatically displayed on the parts used screen, to prompt the keying in of their usage. Where parts are shipped but not used on the job, the available stock in the engineer's stockroom is updated.

The service stockroom, supply stockroom and central warehouse must be set up in a DRP network if you want to run DRP net change processing. If no network is set up, the net change triggers are not set.

The sequence of operations to set up a DRP network covering service, supply and central stockrooms is:

- Set up the calendar for the current year in the Calendar Maintenance option in DRP.
- Set up periods for the central and supply stockrooms using the Period Maintenance option in DRP.
- Set up the central and supply stockrooms using the Stockroom Maintenance option in DRP.
- Set up customers for the central, supply and service stockrooms using the Customer Maintenance option in DRP. The internal customer has to be set up in Accounts Receivable, using the Names And Addresses option.
- Create the network using the Network Maintenance option in DRP, giving the service network a code linked to the central stockroom.
- Set up the planners using the Planner Maintenance option in DRP. Use the central stockroom with user-defined detail for the planner, and set up a planner linked to the supply stockroom.
- Set up items using the Distribution Item Master Maintenance option in DRP.
    - For the central stockroom the source of supply should be set to 1, the order policy should be A, and the local stockroom should be the central stockroom.
    - For the supply stockroom the source of supply should be set to 2, the order policy should be A, and the local stockroom should be the supply stockroom. Use the central stockroom for the centre and the central stockroom.

The sequence of actions for a DRP order for Service Management parts is:

- Amend the delivery name and address, if necessary; **Enter** to fill in details of order lines, and **F8=Update** to complete the DRP order.
- Manually allocate the DRP order, using the Order Amend (Transcriptional) option in SOP.
- Print one pick note at a time, using the Picking Notes option in SOP. Ensure that the 'from' and 'to' order numbers are the same.
- Confirm despatch from the DRP order lines, despatch date and despatch method only, using the Allocation By Order option in SOP. No charges data is relevant.
- Confirm the shipment receipt using the Record Transfer Out option in DRP. Then revert to Service Management and technical reporting.

Miscellaneous Costs

Miscellaneous costs incurred in carrying out the service or repair of each piece of equipment are entered, analysed by job category. For each cost, a cost / charge type code (with its file, or user overwritten description), the cost value and charge value are entered / displayed, or the charge is derived by entering a percentage uplift.

A service job may require more than one engineer visit to complete the work, in which case several technical reports can be entered. The prime report is given the suffix-00, with each subsequent report suffix being incremented by the system. After the final report is entered, the user indicates that the job is finished and fully documented. At this point, the job becomes eligible for pricing and invoicing.

Job Pricing And Invoicing

Service jobs, which are not fully covered under the terms of a service contract, need to be charged to the customer. The pricing and invoicing of jobs is performed after they have been completed through the technical reporting function.

The decision as to which jobs, or elements of jobs, are chargeable is made by referencing the user-defined matrix of contract types and job categories. For the particular combination, the matrix indicates whether labour and / or travel distance and / or travel time and / or miscellaneous costs and / or four types of parts are chargeable to the customer.

Six types of charge can potentially appear on a job invoice:

Contract Visit Fee

Service contracts can include fixed visit charges, to be invoiced after each visit: the contract's billing parameters being set accordingly. Such visits are invoiced through job invoicing at call completion.

The fixed visit charge is obtained from the contract conditions at equipment or header levels, or from the contract rates file.

Labour

The labour charge is calculated from the number of hours booked on the engineer report, multiplied by the hourly rate on the labour price list held for the model. The hourly rate used will be either a contract, or a non-contract, rate depending on whether the piece of equipment was covered by contract when the job was logged. The hourly rate will be uplifted by a premium factor for work performed at overtime rates.

A fixed labour charge may be applied, if required by the job's charge matrix set up in Contract Type / Job Category Maintenance.

Travel Labour

The travel time is calculated from the number of customer travel hours and minutes booked on the engineer's technical report, multiplied by the hourly travel rate from the labour price list held for the model.

A fixed travel labour charge may be applied, if required by the job's charge matrix set up in Contract Type / Job Category Maintenance.

Parts

The charge for parts used by the engineer is calculated according to the price list and discount list defined for the customer site. The price and discount list details are obtained from Sales Order Processing.

Miscellaneous Costs

The charges are based on the values entered through the technical reporting function.

Chargeable Travel

The charges may be based on a standard distance value, the actual distance driven, or zones. Data is set up on the Additional Service Details record for each site.

All jobs, even non-chargeable ones, should be processed through the job pricing and invoicing functions, since the cost values of labour, engineer travel time, miscellaneous costs and parts, for all jobs, are calculated by these routines.

Pending Invoice Line

An invoice line not yet assigned an invoice number, which is waiting to be grouped with other invoice lines to form an invoice.

Description

A batch job is submitted by this menu option to price jobs for the current branch.

Only job lines which have been fully technically reported are processed by this function. This is when **F10** has been taken against any technical report line relating to the job line.

Pending invoice lines are generated for all job lines which have been fully technically reported but not yet invoiced. If a job bill to account exists, this is used to generate the pending invoice. If there is no job bill to account but there is an ownership account, this is used instead. If there is no ownership account, the site account is used.

The price charged for each job line is built up from the following five individual charges:

Labour Charge

Each labour hours entry from each technical report for the job line is priced individually.

A labour charge is only made, if the contract type (of the contract which covered the piece of equipment when the job was created) and job category (entered against the labour hours on the technical report) combination is defined as chargeable for labour, on the cover type / job category file.

Fixed charges: these are applied, prior to any variable labour charges, if defined on the cover type / job category file.

Variable charges: these are applied on the time remaining, after any fixed charges. The labour hours, on each technical report entry, are multiplied by the premium factor (from the codes file LHTY record) for the labour hours type (indicates basic, time and a half, double time etc.).

The result is multiplied by an hourly labour rate from a labour rate price list for the appropriate currency, which is derived as follows:

The labour rate price list is taken from the customer additional details, then the model file. The price list contains four, hourly labour rates: contract, non-contract, workshop and travel. If the engineer is a workshop engineer, that rate is used; otherwise, for work, depending on whether the work has been carried out under a valid contract or not, the contract or non-contract rate is used; and if travel time is involved, on contract or not, the travel rate is used.

In identifying the actual price list record to be used, the pricing date is taken into account, i.e. the price list record effective at the pricing date will be used. The pricing date is determined as follows:

If the equipment is not covered by contract:

- Pricing date=Visit date (from technical report)

If the equipment is covered by contract:

If the Use Start Date Rates field on the contract which covers the equipment is set to on:

- Pricing date=Contract start date

If the Use Start Date Rates field on the contract, which covers the equipment, is set to off, and the job is a planned (scheduled) job, and the visit date is before the price held until date on the contract which covers the equipment, then:

- Pricing date=Contract start date

If not, or if price held until date is zero:

- Pricing date=Visit date

If the job is a 'callout' (breakdown) job and the call logged date is before the price held until date on the contract which covers the equipment, then:

- Pricing date=Contract start date

If not, or if price held until date is zero:

- Pricing date=Call logged date

The labour cost value is always calculated, whether or not the labour is chargeable. The cost value, in base currency, is the number of labour hours multiplied by the engineer cost rate (taken from the engineer master record of the engineer who performed the work). No premium factor is applied to the cost rate.

Parts Charge

A charge for a part is only made if the contract type (of the contract which covered the piece of equipment when the job was created) and job category (entered against the part number on the technical report) combination is defined as chargeable for parts on the cover type / job category file. When reporting parts used on a job, the charge % may be reset from 0% to any value up to 100%.

Parts are priced by retrieving the parts price, discount list, and currency codes held on the Additional Service Details. Using the price and discount list records, defined for these codes in Sales Order Processing (and effective for the pricing date, as derived for the labour pricing calculations), the prices and discounts are calculated in the appropriate prime currency.

The cost value of parts used is always calculated, whether or not the parts are chargeable. Parts costs are always set up in Inventory Item / Stockroom Details in base currency. The cost value is calculated in base currency, not within the Price Jobs function, but when Inventory Management is updated with the detail of the parts used; this takes place when **F9=Complete Technical Report Line**, or **F10=Complete Job Line Report**, is taken within technical reporting.

Miscellaneous Charge

A miscellaneous charge is only made if the contract type (of the contract which covered the piece of equipment when the job was created) and job category (entered against the miscellaneous charge on the technical report) combination is defined as chargeable for miscellaneous charges on the cover type / job category file.

The charge value of each miscellaneous charge line is the charge value entered on the line itself in the specified currency.

The cost value of each miscellaneous charge is always calculated, whether or not the item is chargeable. The cost value is the cost value entered on the line itself in the specified currency. If the specified currency is not the base currency, the system automatically converts the prime currency value to base.

Travel Charge

Fixed charges: these are applied prior to any variable travel charges, if defined on the cover type / job category file.

Variable charges: these are applied on the time remaining after any fixed charges.

The prime currency travel charge is calculated as the billable customer travel time entered on the technical report, multiplied by the travel rate obtained from the labour rates price list. No premium factor is applied to this rate.

The base currency travel cost value is always calculated for engineer travel time. The cost value is the number of travel hours multiplied by the engineer cost rate (taken from the engineer master record of the engineer on the technical report).

Contract Visit Charge

If the job line being invoiced is part of a planned maintenance job (i.e. has been generated via the Load Planned Maintenance Jobs option), and the contract under which the job is performed is currently defined as being invoiced per visit (i.e. Adv / Visit / Arr=1 on the contract's billing parameters), a contract visit charge is generated (unless calculated as zero).

The contract visit charge is calculated as described in the previous section Price Contracts: the number of visits covered by the job is multiplied by the visit fee. The visit fee may default from the contract rates file, or be specified as a contract condition at header and / or equipment line levels.

Engineer Diary

Service Management provides an engineer diary, which enables the planning and scheduling of non-machine-related jobs and other commitments for each engineer. Job appointments are automatically entered in the diary for the engineer, at the time they are made in call logging or in work allocation.

The next five working days (as defined in the daily calendar file) are displayed, starting from the system (current) date. Each day in the diary is free format; bookings can be entered for any date, time and duration, or for a date and time 'from' with a date and time 'to'.

Diary entries

Diary entries are made as required and are used to reserve the times for events such as holidays, training courses, sickness, dental appointments, etc. As entries for engineers are scheduled for a date and time, they will appear both in the engineer's work queue and in the graphical display of his jobs.

The diary is accessible from the Engineer Work Allocation routine.

Job Escalation

Service Management can alert you to any service jobs that are approaching their response time and are still outstanding. Job escalation runs so long as the Service Management sub-system is active.

Note: The sub-system library list must be tailored to take account of:

- the use of consolidated files' libraries in place of IN, OE, SL and GL files' libraries.

- the version of System Manager in use, for adjusting the four IPG libraries.

All service jobs created in the system will be stamped with a target response date and time, which is calculated as follows:

For call out jobs (i.e. those entered to the system using the Job Logging / Maintenance function), the target response date and time is calculated as:

Date / time call logged + response hours

... where response hours is derived from the special serial numbers file, or the contract which covers the piece of equipment if a contract exists. If the equipment is not covered by contract, the response hours default to the value for the contract type *NO. Manual overrides of the response hours may be made, if special circumstances arise.

In calculating target response date / time, the system takes into account working / non-working days, and the standard service 'window' for the company, i.e. the times of the day between which service is provided. These are all user-defined.

For planned jobs (i.e. those generated from the service schedules defined for contracts), the target response date and time is calculated as:

Target date=Date of last working day of the service period in which the job is scheduled

Target time=Start time of the working day

You define the parameters which determine the escalation time fence, the interval between each escalation step, and the users who are to be alerted at each step. The system monitors each outstanding job, comparing its target date / time with the current system date / time. Once the job enters the defined escalation time fence, it will start to escalate and will have its escalation step recalculated at frequent intervals: the escalation step moving from high to low value as the job approaches its target date / time. Escalation step zero denotes that the job has reached (or gone beyond) its target time.

Note: The system assumes a 7-day working week. If less than this is actually worked, the system still escalates what would be Monday's work on a Sunday, not on the preceding Friday.

As a job moves from one escalation step to the next, the system can send a message to pre-defined users to alert them to the fact that the job has been escalated to the next level, thereby allowing them to take appropriate action.

Jobs will be removed from the escalation procedure once they reach a certain status. This is user-defined within the job line status code record; for example, escalation may be required to stop, once the job is allocated to an engineer, or once the engineer has arrived and started work. A screen enquiry allows the escalation status of jobs to be viewed easily, using various selection parameters.

Example of Call Escalation

Escalation Step	Reporting Times - (1 Hour Intervals)	
0	16:00	Target Time
1	15:00	
2	14:00	
3	13:00	
4	12:00	Escalation Time Fence (4 hours) starts

In the above example, the escalation time fence is set to 4 hours and the reporting interval to 1 hour.

A job with target time of 16:00 will start to escalate at 12:00.

Between 12:00 and 12:59, the escalation step will be calculated as 4, because the job is within 4 hours of target.

Between 13:00 and 13:59, the escalation step will be calculated as 3, because the job is within 3 hours of target, etc.

At 16:00, or later, the escalation step will be calculated as 0, because the job has reached or passed its target time.

Remote Communications Links to and from Engineers

Outgoing and Incoming job processing records can be held on files interfaced to a communications gateway. Records are validated and may be corrected centrally, or returned to the appropriate engineers for correction.

A variety of carriers and mobile terminals can be integrated with Service Management. Users are able to select their own preferred route for the gateway, the communications link and the remote devices.

Whether or not the remote communications link is active, normal job processing routines continue without any apparent change.

Time Zones

This function enables you to operate across different time zones. The zones are specified as a number of hours difference (called the 'offset time') from the system time. You can specify zones at branch level and at customer additional details level.

Once you have specified the offset time, when service calls are received the customer's date and time is recorded with them, and are used to calculate target times.

The time zone offset times are set up in Inventory, taking the Descriptions option from the Maintenance menu and adding times under the major type TIMZ.

Enquiries

Service Management provides a comprehensive range of enquiries for:

Master Files

These generally show the same details as their counterpart maintenance routines.

Transaction History

A job history enquiry is available for viewing details of historical service work performed against equipment. An invoice enquiry is available for viewing details of a service invoice, whether it be a contract, job, or sundry invoice.

Status

An escalation enquiry provides up-to-date information on the escalation status of service jobs.

Loading Service Management Data

The following data must be loaded for each service company which is to be processed within the live module.

The data is divided into two groups:

- Basic data - data used mainly for reference purposes, which is low in volume and not subject to ongoing change, e.g. codes.
- Major data - data which constitutes the bulk of the module and is subject to regular change, e.g. customers, contracts, etc.

Some of the data listed below is marked optional. This means that, in system terms, the data is not essential to running the module; but note that, in practice, the data may well be essential, e.g. contracts.

Basic (Parameter) Data

- Company profile
- System parameters file
- Branch records
    - one for each service branch
- User / branch authority records
    - one for each user and branch (to which the user is authorised) combination
- Codes / parameter file records.
    - System-required parameters are identified by * in the first position of the parameter type description: these must be set up, or copied from the demonstration data supplied. In addition, set up appropriate code values for each of the following types:
    - ABSC (Engineer absent)
    - CCRC (Contract cancellation reasons)
    - CHGT (Miscellaneous costs / charges)
    - CORA (Corrective actions)
    - DAYT (Day types)
    - EGRD (Engineer grades)
    - FLTC (Faults)
    - LHTY (Labour hours / overtime types)
    - REGN (Service regions)
    - RRES (Credit release reason)
    - STAT (Equipment status)
    - TXTO (Text destinations)
- Calendar control record
- Daily calendar file (& rebuild period end dates)
    - for all years over which current service contracts are spread
- Division records
    - one for each service division
- Model group / sub-group records
    - one for each model group / sub-group combination
- Geocode / territory records (optional)

  - one for each service territory, with a list of geocodes (postal codes) within the territory

- Field service group
  - one for each group or equipment (division / model family / model to be serviced)
- Team
  - one for each engineer team
- District
  - one for each work controller, to authorise work on teams
- Engineer records
  - one for each service engineer
- Labour rates price list records
  - one for each set of labour rates to be linked with models (and for different currencies where World Trade and Multi-Currency are active)
- Zone charge (optional)
  - one for each zone (and currency where World Trade and Multi-Currency are active)
- Model records
  - one for each model code
- Volume segment
  - minimum one, optionally more for each model, or group of models
- Job category records
  - one for each job category code
- Scheduled visit profile records (optional)
  - one for each standard schedule of planned maintenance visits
- Contract type records
  - one for each contract type code
- Contract type / job category records
  - one for each valid combination of contract type, job category, (and currency where World Trade and multi-currency are active) with invoicing indicators
- Contract rates records (optional)
- Escalation parameters (optional)
- Tax codes (for World Trade and multi-currency only.)
  - for each country's contract type / job category combinations
- Inventory descriptions file

- DEPT (Departments)
  SS - Service workshop
  S3 - Service
- MANT (Mantle groupings)
  A - 'Attack' customers
  C - Central Govt. Depts.
  L - Local Govt. Depts.
  M - Multi-nationals
  S - 'Special' customers
- MFID (Manufacturer's ID)
  * - Mfr not specified
  BRIT - Briton Ferry Steel Co Ltd
  EAGL - Eaglesbush Nuts& Bolts
  FALC - Falcon Electrical
  GRU - Herbert Grudgings (Spares)
  JONE - Jones Britannia Foundry Ltd
  ROKI - Rock Island Arsenal (NJ) Ltd
  STEP - Steptoe & Son Precision Engrs
  TAY - Taylor Engineers (Rothley)
  WRAY - Wray Optical Co Ltd
- OTHR (Customer search parameters)
  END1 - Cust account no. 9
  END2 - Phone no.332
  START1 - Cust account no. 2
  START2 - Phone no. 313
- PTYP (Item type)
  4 codes are essential to categorise parts:
  C - Consumables
  J - Major spares
  N - Minor spares
  S - Supply items
  2 codes assist analysis of machine sales:
  M - Machine sale via SOP
  P - Peripheral sale via SOP
- SPED (Urgency indicator)
  1 - Special courier: ETA TBA
  2 - By noon tomorrow: promised
  3 - By noon tomorrow: guaranteed
  4 - 24 hour delivery: guaranteed
  5 - By 17:30 today
  6 - By 22:00 Monday
- STKU (Stockroom usage code)
  B - Service spares stockroom
  W - Service workshop: company stock
  X - Service workshop: customer stock
- TIMZ (Time zones)
  E01 - East + 1 hour
  E02 - East + 2 hours
  E03 - East + 3 hours

W01 - West - 1 hour
W02 - West - 2 hours
W03 - West - 3 hours

Major Data

- Customer account records
  - one for each customer account.
- Installation address records
  - one for each site address, within a customer account.
- Customer additional details (Service Management) records
  - one for each customer site.
- Installation records
  - for each customer site, details of each piece of equipment at the site.
- Contract records (optional)
  - one for each active service contract, with details of each piece of equipment covered and service schedules.